

Unit-V: Display Devices: LCD, LED, Seven Segment Display, Alphanumeric Display, Electronic Ammeter and Voltmeter, Digital Multi-meter, Cathode Ray Oscilloscope (CRO), Digital Storage Oscilloscope (DSO) Give me proper extended notes of this Without missing anything Give it to me in pdf form

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Response: UNIT-V: Display Devices, Electronic Meters, and Oscilloscopes

1. Display Devices

Display devices convert electrical signals into visual information, categorized into:

- **Active (Emissive):** Emit light directly (e.g., LEDs).
- **Passive (Non-Emissive):** Modulate ambient or backlight illumination (e.g., LCDs).

1.1 Light Emitting Diode (LED)

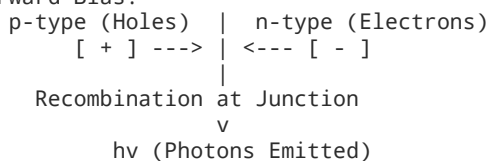
Working Principle

An LED is a heavily doped *p-n* junction diode operated under **forward bias**. It functions via **electroluminescence**:

1. Under forward bias, majority carriers (electrons from the *n*-region and holes from the *p*-region) cross the junction.
2. In the depletion region and near the junction boundaries, these excess minority carriers recombine.
3. As an electron in the conduction band recombines with a hole in the valence band, it falls across the bandgap energy (E_g).
4. The excess energy is released as a photon whose wavelength (λ) is governed by:

$$\lambda = \frac{hc}{E_g} \approx \frac{1.24}{E_g \text{ (in eV)}} \mu\text{m}$$

Forward Bias:



Semiconductor Materials

Direct bandgap semiconductors are required so that carrier transitions conserve crystal momentum without requiring phonon interactions:

- **Gallium Arsenide (GaAs):** Infrared ($E_g \approx 1.43$ eV)
- **Gallium Arsenide Phosphide ($\text{GaAs}_{1-x}\text{P}_x$):** Red to yellow depending on x
- **Gallium Phosphide (GaP):** Red or green (doped with Nitrogen)
- **Indium Gallium Nitride (InGaN):** High-brightness blue and green
- **White LEDs:** Blue InGaN chip coated with yellow cerium-doped YAG phosphor ($\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$)

Constructional Details

- **Die Mounting:** The semiconductor chip is mounted on a reflective cup (anode/cathode lead frame) to focus the light output upward.
- **Encapsulation:** Enclosed in transparent or colored epoxy resin that serves both as a protective casing and a focusing hemispherical lens.
- **Current-Limiting Resistor (R_s):**

$$R_s = \frac{V_{CC} - V_D}{I_D}$$

Typical operating parameters: $V_D \approx 1.8 \text{ V}$ to 3.3 V , $I_D \approx 10 \text{ mA}$ to 20 mA .

Characteristics & Parameters

- **V-I Characteristics:** Similar to standard diodes, but with a higher knee voltage (1.8 V for red, up to 3.5 V for blue/white) and lower reverse breakdown voltage ($V_{BR} \approx 3 \text{ V}$ to 5 V).
- **Luminous Intensity vs. Current:** Roughly linear up to the rated saturation current.

Advantages & Applications

- **Advantages:** Solid-state durability (no filament), fast switching speed ($< 10 \text{ ns}$), low power consumption, long operational lifespan ($> 50,000$ hours).
- **Applications:** Status indicators, optical fiber communication sources, dot-matrix screens, traffic signals.

1.2 Liquid Crystal Display (LCD)

Working Principle

Liquid crystals represent a state of matter (mesophase) exhibiting properties between crystalline solids and isotropic liquids. LCDs rely on the **nematic phase**, where elongated rod-like organic molecules maintain parallel directional alignment.

LCDs are **passive displays**—they do not generate light. Instead, they modulate polarized ambient or backlight illumination via the **Twisted Nematic (TN) effect**.

State A: OFF (No Electric Field Applied)

[Polarizer 1 (0°)] $\rightarrow$ [Liquid Crystal Molecule 90° Twist] $\rightarrow$ [Polarizer 2 (90°)] $\rightarrow$ Light Passes (BRIGHT)

State B: ON (Electric Field Applied)

[Polarizer 1 (0°)] $\rightarrow$ [Molecules Untwist & Align to Field] $\rightarrow$ [Polarizer 2 (90°)] $\rightarrow$ Light Blocked (DARK)

Layer Construction (Sandwich Structure)

1. **Vertical Linear Polarizer (0°):** Polarizes incoming unpolarized light.
2. **Glass Substrate with Indium Tin Oxide (ITO) Electrodes:** ITO forms a transparent conductive pattern.
3. **Polyimide Alignment Layer:** Micro-grooved to orient the liquid crystal molecules at the surfaces.
4. **Nematic Liquid Crystal Layer:** Molecules naturally twist by 90° across the gap (5 to $10 \mu\text{m}$).
5. **Horizontal Linear Polarizer (90°):** Oriented orthogonal to the front polarizer.
6. **Reflective Plate or Backlight (CCFL/LED):** Directs light through the assembly.

Electro-Optic Operation

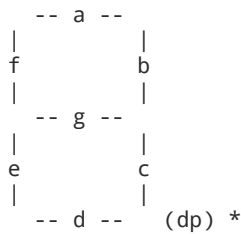
- **OFF State (Zero Field):** Light passes Polarizer 1, is rotated 90° as it follows the molecular twist, and exits Polarizer 2. The segment appears bright.
- **ON State (Electric Field Applied via AC voltage):** The electric dipole moment forces the molecules to align parallel to the field, destroying the 90° optical twist. The polarized light reaches Polarizer 2 unchanged, cannot pass, and is absorbed. The segment appears dark.
- *Note:* LCDs must be driven by an AC square wave (30 Hz to 100 Hz) to prevent irreversible electrochemical degradation (electrolysis) caused by DC currents.

LED vs. LCD

Parameter	LED	LCD
Emission Type	Active (emits light)	Passive (modulates external light)
Power Consumption	Moderate (10 to 50 mW per element)	Extremely low (μW range)
Response Time	Nanoseconds ($< 10 \text{ ns}$)	Milliseconds (1 to 10 ms)
Contrast / Visibility	High contrast, visible in darkness	Needs backlighting in darkness; excellent in sunlight
Viewing Angle	Wide (up to 160°)	Narrow to moderate (contrast shifts off-axis)

1.3 Seven Segment Display (SSD)

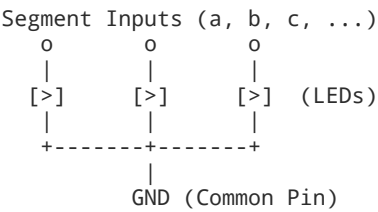
A seven segment display arranges eight LEDs (seven segments labeled *a* through *g*, plus a decimal point *dp*) in a figure-eight configuration to display decimal digits (0–9) and select hexadecimal characters.



Configurations

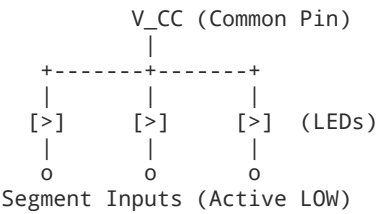
1. Common Cathode (CC)

- All LED cathodes are tied together to system ground (0 V).
- Segments illuminate when driven with a **Logic HIGH (1)** signal through a current-limiting resistor.



2. Common Anode (CA)

- All LED anodes are tied together to the positive rail (V_{CC}).
- Segments illuminate when driven with a **Logic LOW (0)** signal (current sinks into the driver).



Truth Table (BCD to 7-Segment Mapping)

Digit	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

Note: For Common Anode configurations, invert all bit levels (0 → 1, 1 → 0).

Driving Methods

- **Direct Drive:** Dedicated BCD-to-7-Segment decoders/drivers (e.g., 74LS47 for CA, 74LS48 or CD4511 for CC) drive every segment directly.
- **Multiplexed Drive:** The segment lines of multiple digits are wired in parallel; digit common pins are enabled sequentially via high-frequency scanning (> 50 Hz) to prevent visible flicker using the persistence of vision principle.

1.4 Alphanumeric Display

Standard 7-segment displays cannot represent complex characters (e.g., *M*, *W*, *K*, *X*, *Z*). Alphanumeric displays resolve this with additional segments or matrix arrangements.

Types

1. Fourteen-Segment (Starburst) & Sixteen-Segment Displays

- Splits horizontal lines and adds diagonal and vertical central segments.
- Can render all 26 uppercase English letters, digits 0–9, and common punctuation symbols.

Sixteen-Segment Layout:

```
--a1-- --a2--
| \      |      / |
f  h      j      k b
|  \      |      /  |
--g1-- --g2--
| /       |       \ |
e  m      n      p  c
| /       |       \ |
--d1-- --d2--
```

2. Dot Matrix Display (5×7 and 8×8)

- Organizes LEDs or LCD elements into an array of rows and columns.
- A 5×7 matrix (35 dots) can display uppercase and lowercase letters, custom symbols, and basic graphics.
- **Control Architecture:** Managed via row-column scanning. A Character Generator ROM (such as ASCII table lookups) selects column codes while scanning through rows using shift registers.

2. Electronic Meters

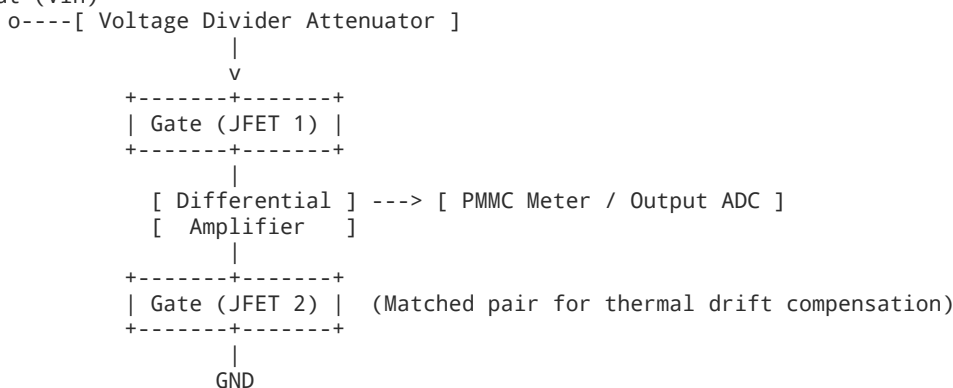
Unlike electromechanical meters (e.g., PMMC), electronic meters feature high input impedance, wide frequency response, and active signal conditioning.

2.1 Electronic Voltmeter and Ammeter

Electronic DC Voltmeter (FET Input Type)

To eliminate circuit loading effects, an electronic voltmeter requires an exceptionally high input impedance ($> 10 \text{ M}\Omega$).

Input (V_{in})



- **Attenuator Network:** High-value metal film resistors divide down input voltages to safe operational levels.
- **Balanced JFET Differential Pair:** Provides an input resistance of 10^9 to $10^{12} \Omega$. A balanced bridge layout cancels out drift caused by temperature variations and supply voltage fluctuations.
- **PMMC/ADC Indicator:** Reads the bridge unbalance current, which remains proportional to V_{in} .

Electronic AC Voltmeter

AC signals are conditioned, amplified, and rectified to produce a measurable DC equivalent:

1. **Average-Responding:** Uses full-wave rectifiers calibrated to display the RMS value for a pure sine wave ($V_{\text{RMS}} = 1.11 \times V_{\text{avg}}$). Significant waveform errors occur with non-sinusoidal inputs.
2. **Peak-Responding:** Uses diode-capacitor peak detector circuits ($V_{\text{peak}} = 2 \cdot V_{\text{RMS}}$ for sine waves).

3. **True RMS Voltmeter:** Uses analog multipliers, squaring, and low-pass filtering modules:

$$V_{\text{RMS}} = \frac{1}{T} \int_0^T v^2(t) dt$$

Accurately measures complex, distorted, or pulsed non-sinusoidal waveforms.

Electronic Ammeter

Measures current by converting it into a proportional voltage across a low-resistance current shunt (R_{sh}):

- Direct current causes a small drop: $V_s = I \cdot R_{sh}$.
- **Op-Amp Transimpedance Configuration:** For small currents (picoamperes to microamperes), an op-amp transimpedance amplifier drives the non-inverting terminal to virtual ground, converting I_{in} directly into an output voltage:

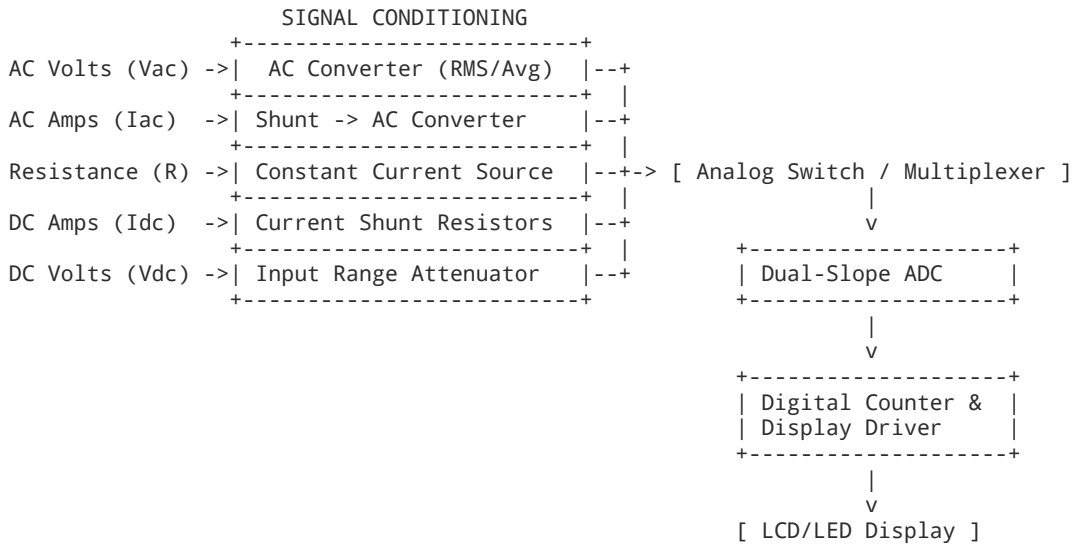
$$V_{out} = -I_{in} \cdot R_f$$

This virtual ground eliminates the meter's insertion voltage drop ("burden voltage").

2.2 Digital Multimeter (DMM)

A DMM measures DC voltage, AC voltage, DC current, AC current, and resistance, digitizing the values for clear numeric readout without parallax errors.

Architectural Block Diagram



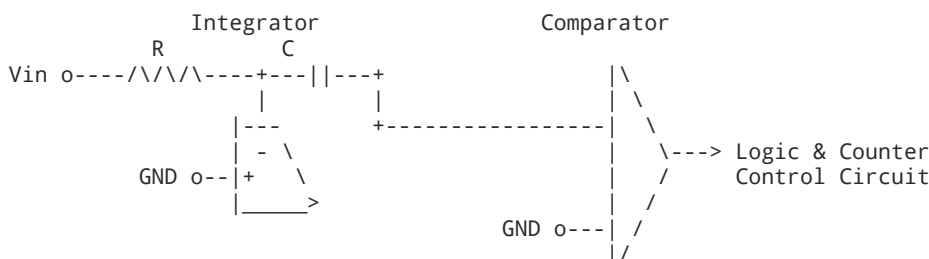
Signal Conditioning Blocks

1. **DC Voltage:** A compensated resistive ladder attenuator reduces high voltages to the core ADC range (± 200 mV or ± 2 V).
2. **Resistance Measurement:** A precision constant current generator (I_{ref}) passes through the unknown resistor R_x . The measured voltage drop is directly proportional to the resistance:

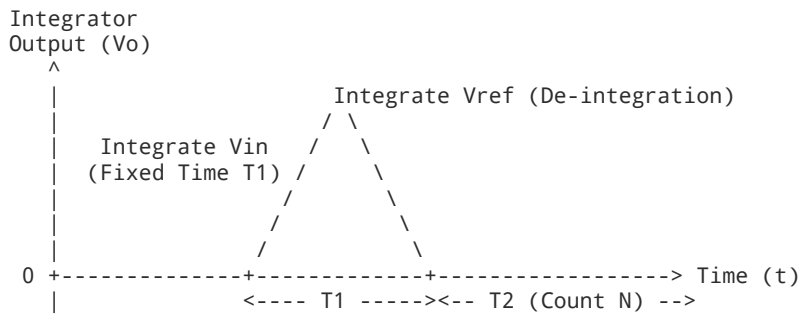
$$V_x = I_{ref} \cdot R_x$$
3. **Current Measurement:** Multi-range shunt resistors (R_{sh}) generate an internal voltage drop ($I \cdot R_{sh}$).
4. **AC Measurement:** True RMS or average-responding rectifier converts AC inputs to proportional DC voltages.

Core Converter: Dual-Slope Integrating ADC

The dual-slope integrating ADC is favored in DMMs for its linear accuracy and high rejection of power-line noise (50 Hz/60 Hz hum).



Operating Cycle



- **Phase 1: Signal Run-Up (Fixed Time T_1):** The integrator connects to the unknown voltage V_{in} for a fixed duration T_1 , corresponding to a full counter cycle (N_1 counts):

$$V_o = -\frac{1}{RC} \int_0^{T_1} V_{in} dt = -\frac{V_{in} \cdot T_1}{RC}$$

- **Phase 2: Reference Run-Down (Measured Time T_2):** The integrator switches to a precision reference voltage ($-V_{ref}$, opposing polarity). The integrator ramps back down toward 0 V. The counter records the clock pulses N_2 required for the output to return to zero:

$$V_o = 0 \implies \frac{V_{in} \cdot T_1}{RC} = \frac{V_{ref} \cdot T_2}{RC}$$

- **Result:**

$$T_2 = T_1 \cdot \left(\frac{V_{in}}{V_{ref}} \right) \implies N_2 = N_1 \cdot \left(\frac{V_{in}}{V_{ref}} \right)$$

Key Takeaway: The result is independent of passive component tolerances (R and C) and clock frequency drift, as both affect T_1 and T_2 equally.

Specifications

- **Digit Counts ($3\frac{1}{2}$, $4\frac{1}{2}$ digits):**
 - The "1/2" digit represents the most significant digit (MSD). It can only read 0 or 1.
 - A $3\frac{1}{2}$ digit display presents a maximum count of **1999** (ranging 0000 to 1999).
 - A $4\frac{1}{2}$ digit display presents a maximum count of **19999**.
- **Resolution:** Smallest increment the meter can display on its lowest range:

$$\text{Resolution} = \frac{\text{Full Scale Range}}{\text{Maximum Count}}$$

- **Sensitivity:** Smallest change in input signal the instrument can reliably detect.

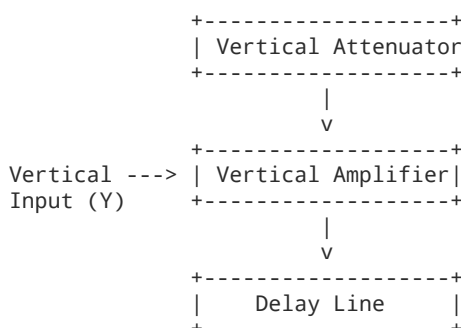
3. Oscilloscopes

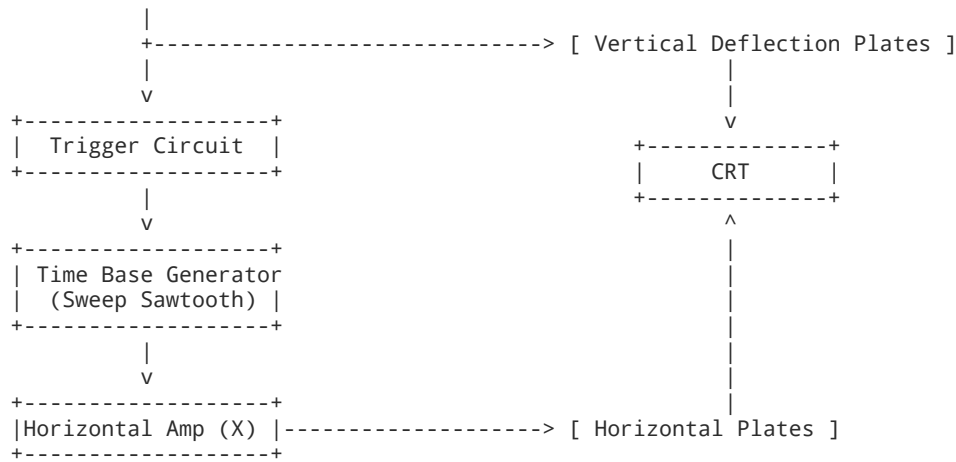
Oscilloscopes plot voltage against time, providing a visual profile of dynamic electrical signals.

3.1 Cathode Ray Oscilloscope (CRO)

A conventional analog CRO uses electrostatic deflection to steer an electron beam across a phosphor-coated screen in real time.

Block Diagram of an Analog CRO



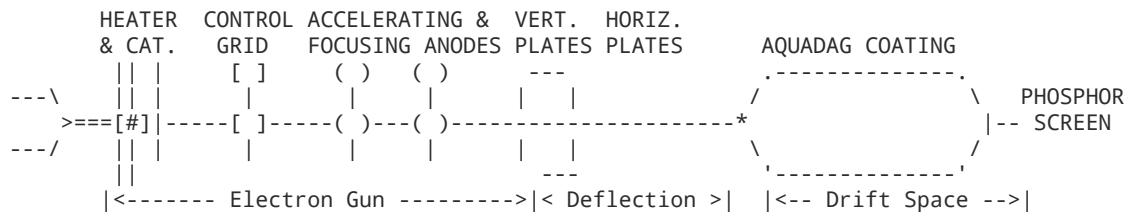


Functional Subsystems

- Vertical Amplifier:** Amplifies low-level input signals to voltages suitable for deflection plates (20 to 100 V). Sets the instrument's bandwidth and vertical deflection factor (Volts/Div).
- Delay Line:** Introduces a fixed delay of 100 to 200 ns to the vertical signal. This allows the trigger circuits and sweep generator to stabilize before the beam reaches the vertical plates, preventing the leading edge of the waveform from being cut off.
- Trigger Circuit:** Samples the input waveform and produces synchronized pulses at precise threshold crossings, stabilizing recurring waveforms on screen.
- Time Base Generator (Sweep Circuit):** Produces a linear sawtooth voltage. The linear rise sweeps the electron beam horizontally across the screen at a constant speed, while the rapid fall-time (flyback) returns the beam to the left edge.
- Horizontal Amplifier:** Amplifies the sawtooth sweep voltage or an external horizontal input signal (for X-Y operation).
- High-Voltage and Low-Voltage Power Supplies:**
 - HV:** Supplies -1 to -3 kV for the CRT electron gun and $+10$ to $+20$ kV post-deflection acceleration.
 - LV:** Powers the internal transistorized amplifiers, sweep circuits, and display logic.

Cathode Ray Tube (CRT) Details

The CRT is an evacuated envelope containing an electron gun, deflection assembly, and a phosphor screen.



1. Electron Gun

- Indirectly Heated Cathode:** Barium and strontium oxide coated cap that emits electrons via thermionic emission.
- Control Grid:** A cup-shaped cylinder biased at a negative potential relative to the cathode. Adjusting this negative bias varies the electron flow, controlling trace **Intensity**.
- Pre-Accelerating and Accelerating Anodes:** Biased at high positive potentials ($+1$ to $+3$ kV) to impart kinetic energy to the electrons.
- Focusing Anode:** Positioned between the accelerating anodes. By adjusting the electrostatic lens formed between these fields, the beam converges to a sharp focal point on the screen (**Focus Control**).

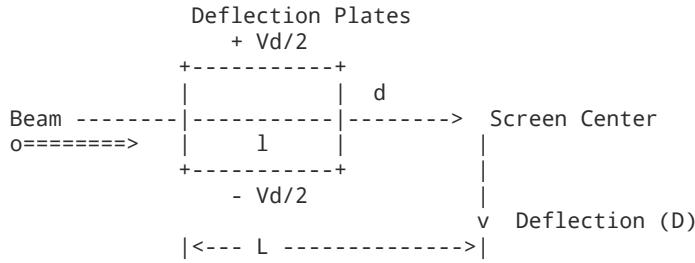
2. Deflection System

- Electrostatic deflection plates are arranged in pairs: **Vertical Plates (Y)** deflect the beam up and down; **Horizontal Plates (X)** sweep it left to right.

3. Screen and Coatings

- Phosphor Coating:** Converts the kinetic energy of striking electrons into visible light (fluorescence during impact, phosphorescence after impact). Typical material: P31 phosphor (zinc sulfide doped with copper), emitting a green trace with balanced persistence.
- Aquadag Coating:** An electrically conductive aqueous solution of graphite painted on the inner glass walls. It collects secondary emission electrons knocked loose from the phosphor screen, returning them to ground and preventing charge buildup on the glass.

Derivation: Electrostatic Deflection Sensitivity



Let:

- e = Electron charge (1.602×10^{-19} C)
- m = Electron mass (9.109×10^{-31} kg)
- V_a = Accelerating anode voltage
- V_d = Deflection voltage applied between plates
- l = Axial length of deflection plates
- d = Separation distance between plates
- L = Distance from the center of the deflection plates to the screen
- v_{ox} = Axial entry velocity of the electron

Step 1: Entry Velocity

The kinetic energy imparted to the electron by the accelerating potential equals the electrical work done:

$$\frac{1}{2}mv_{ox}^2 = eV_a \implies v_{ox}^2 = \frac{2eV_a}{m}$$

Step 2: Acceleration Within Deflection Plates

The uniform electric field between the plates is $\mathcal{E}_y = \frac{V_d}{d}$.

The upward electrostatic force acting on the electron is $F_y = e\mathcal{E}_y = \frac{eV_d}{d}$.

By Newton's second law, vertical acceleration is:

$$a_y = \frac{F_y}{m} = \frac{eV_d}{md}$$

Step 3: Transit Time and Vertical Velocity at Exit

Transit time through the plates of length l :

$$t = \frac{l}{v_{ox}}$$

The vertical velocity acquired upon exiting the plates is:

$$v_y = a_y \cdot t = \left(\frac{eV_d}{md} \right) \left(\frac{l}{v_{ox}} \right)$$

Step 4: Total Deflection on Screen (D)

The deflection angle θ is:

$$\tan \theta = \frac{v_y}{v_{ox}} = \frac{eV_d l}{mdv_{ox}^2}$$

Substituting $v_{ox}^2 = \frac{2eV_a}{m}$:

$$\tan \theta = \frac{eV_d l}{md \left(\frac{2eV_a}{m} \right)} = \frac{lV_d}{2dV_a}$$

Because deflection angle θ is small, the beam appears to emerge from the plate midpoint. Over distance L :

$$D = L \cdot \tan \theta = \frac{LlV_d}{2dV_a}$$

Step 5: Deflection Factor & Deflection Sensitivity

- **Deflection Sensitivity (S):** Deflection per unit deflection voltage:

$$S = \frac{D}{V_d} = \frac{Ll}{2dV_a} \quad (\text{expressed in mm/V or cm/V})$$

- **Deflection Factor (G):** Inverse of deflection sensitivity:

$$G = \frac{1}{S} = \frac{2dV_a}{Ll} \quad (\text{expressed in V/mm or V/cm})$$

Lissajous Figures

When sinusoidal voltages of identical or harmonically related frequencies are applied simultaneously to both the horizontal (X) and vertical (Y) plates, the resulting screen patterns are **Lissajous figures**.

Phase Shifts ($f_x = f_y$, Equal Amplitudes):

0° Line 45° Ellipse 90° Circle 135° Ellipse 180° Line

1. Phase Measurement ($\Delta\phi$ for $f_x = f_y$)

For an ellipse centered on the axes:

$$\sin \phi = \frac{Y_1}{Y_2} \Rightarrow \phi = \arcsin \left(\frac{Y_1}{Y_2} \right)$$

- Y_1 = Y-axis intercept (where the ellipse crosses the vertical center axis)
- Y_2 = Maximum vertical deflection (Y_{max})

2. Frequency Ratio Determination (f_y/f_x)

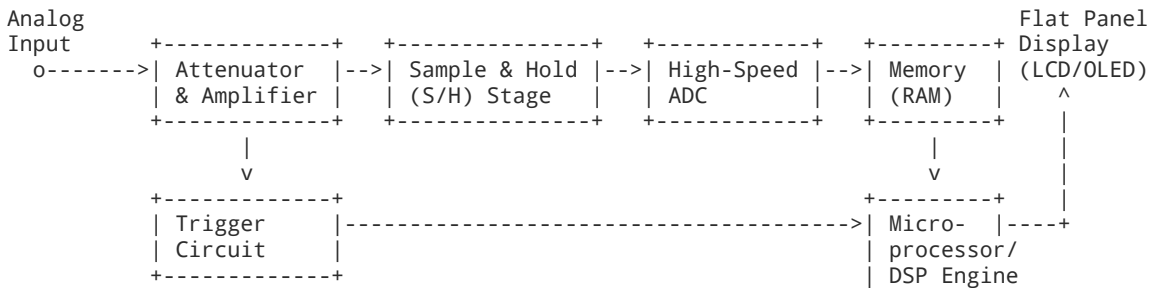
By drawing horizontal and vertical tangent lines across the display:

$$\frac{f_y}{f_x} = \frac{\text{Number of points where horizontal line is tangent to loops}}{\text{Number of points where vertical line is tangent to loops}} = \frac{N_H}{N_V}$$

3.2 Digital Storage Oscilloscope (DSO)

A DSO digitizes incoming analog signals with an analog-to-digital converter (ADC), stores the binary data in RAM, and reconstructs the waveform on a flat raster display.

Architectural Block Diagram



Signal Processing Stages

1. **Signal Conditioning:** Wideband attenuators and pre-amplifiers scale input levels to match the dynamic range of the internal ADC.
2. **Sample and Hold (S/H):** Freezes the rapidly changing analog input voltage during conversion windows to prevent aperture jitter.
3. **ADC Digitization:** Flash ADCs convert analog voltages into high-speed 8-bit to 12-bit parallel digital words at gigasample-per-second (GSa/s) rates.
4. **Acquisition Memory:** Circular RAM buffers store consecutive data points.
5. **Microprocessor / DSP Processing:** Computes automated metrics (Rise time, Peak-to-Peak, RMS, Frequency), executes interpolation algorithms, or performs Fast Fourier Transform (FFT) analysis.

6. **Display System:** Draws the reconstructed waveform onto an active matrix LCD using vector generation or raster scan techniques.

Sampling Techniques

1. Real-Time Sampling

- The ADC captures all waveform points during a **single trigger pass**.
- Essential for transient, single-shot, or non-repetitive phenomena.
- Governed strictly by the **Nyquist-Shannon Sampling Theorem**:

$$f_s \geq 2 \cdot f_{max}$$

(In practice, oscilloscopes require $f_s \geq 4 \cdot f_{max}$ to $10 \cdot f_{max}$ for accurate waveform reconstruction).

2. Equivalent-Time Sampling

Used for repetitive signals whose bandwidth exceeds the ADC's maximum real-time sampling rate.

- **Sequential Sampling:** Samples one point per cycle at systematically delayed time intervals across successive cycles.
- **Random Interleaved Sampling (RIS):** Samples continuously at an internal clock rate asynchronous to the input signal, using high-resolution time-to-digital converters to compute each sample's offset relative to the trigger.

3.3 Comparative Analysis: CRO vs. DSO

Feature	Cathode Ray Oscilloscope (CRO)	Digital Storage Oscilloscope (DSO)
Operating Principle	Direct deflection of an electron beam onto a phosphor screen	Digitization, storage in solid-state RAM, and numerical display
Display Device	Vacuum Cathode Ray Tube (bulky, fragile)	Flat-panel LCD/OLED (compact, lightweight)
Transient Capture	Poor; cannot store single-shot events (fades with phosphor persistence)	Excellent; single events remain stored in memory indefinitely
Pre-Trigger Viewing	Impossible; requires external triggers and delays	Standard feature; the circular memory buffer preserves pre-trigger samples
Data Processing	None; readings are manual using graticules	Automated math functions (FFT, RMS, Peak-to-Peak, cursor metrics)
Storage Capacity	Transient only (determined by phosphor persistence)	Infinite duration storage; files can be exported via USB, LAN, or GPIB
Cost & Complexity	Simple circuitry, but higher component weight and volume	Higher digital complexity, but lower assembly weight and production cost

Quick Reference: Primary Formulas

- **Photon Emission (LED):**

$$\lambda = \frac{hc}{E_g} \approx \frac{1.24}{E_g \text{ (eV)}} \mu\text{m}$$

- **Current-Limiting Resistor:**

$$R_s = \frac{V_{CC} - V_D}{I_D}$$

- **Dual-Slope ADC Conversion:**

$$V_{in} = V_{ref} \cdot \left(\frac{N_2}{N_1} \right)$$

- **Electrostatic Deflection:**

$$D = \frac{L \cdot l \cdot V_d}{2 \cdot d \cdot V_a}$$

- **Deflection Sensitivity:**

$$S = \frac{D}{V_d} = \frac{L \cdot l}{2 \cdot d \cdot V_a}$$

- **Phase Shift via Lissajous Ellipse:**

$$\phi = \arcsin\left(\frac{Y_1}{Y_2}\right)$$

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